

SUMMER INTERNSHIP REPORT

Customer Behavior Analysis

B.TECH SEMESTER–VII



Computer Department

**BIRLA VISHVAKARMA MAHAVIDYALYA
ENGINEERING COLLAGE**

(An Autonomous Institution)

AY: 2025-26, Semester VII

Name: Trushti Rupapara

Id Number: 23CP019

Index

1. Abstract
2. Discription of the project
3. Introduction
4. Problem statement ,Solution
5. Project Workflow ,Data cleaning
6. EDA
7. Dashboard
8. Insights

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ms. Trushti Rupapara, has successfully completed a six-week Internship in the field of Data Analytics, Starting from 15 May 2026 to 15 June 2026.

During her Internship tenure with us, She was exposed to various aspects of Data Analysis, including Data Preprocessing, Visualisation, Exploratory Data Analysis, and the use of Python libraries such as NumPy, Pandas, and Matplotlib.

This is to further certify that Ms. Trushti Rupapara has successfully completed all the assigned tasks to the best of her abilities.

We wish all the success in her career.

Yours truly,

Nehal

**Ms. Nehal Kankotiya,
(HR Manager)
SEOFEB, Ahmedabad**

Abstract

Customer behavior analysis is a critical business function that helps organizations understand purchasing patterns, customer preferences, loyalty behavior, and revenue drivers. The objective of this project was to develop a complete end-to-end data analytics solution that transforms raw customer shopping data into actionable business insights.

The project follows a corporate-style analytics workflow consisting of:

- Data Collection from CSV dataset
- Data Cleaning & Transformation using Python
- Data Loading into PostgreSQL Database
- Business Analysis using SQL Queries
- Dashboard Development using Power BI
- Strategic Business Recommendations

The final outcome is an interactive dashboard that enables stakeholders to analyze customer segments, purchasing trends, subscription behavior, shipping preferences, and revenue contributions across different customer groups.

Description of the Project

Customer behavior analysis is one of the most important aspects of modern business analytics. Organizations collect large amounts of customer data but often struggle to convert it into actionable insights.

The purpose of this project is to simulate an industry-level analytics workflow by performing:

Data Collection

Customer shopping behavior dataset containing customer demographics and purchase information.

Data Cleaning

Using Python and Pandas:

- Missing value handling
- Data transformation
- Data standardization
- Feature engineering

Database Integration

Loading cleaned data into PostgreSQL for business analytics.

SQL Analysis

Writing business queries to analyze:

- Customer Segments
- Revenue Trends
- Subscription Impact
- Product Performance
- Purchase Behavior

Dashboard Development

Building an interactive Power BI dashboard to visualize customer behavior and business KPIs.

TECHNOLOGIES USED :

Programming Language

- Python

Libraries

- Pandas
- NumPy
- SQLAlchemy
- Psycopg2

Database

- PostgreSQL
- pgAdmin 4

Visualization Tool

- Power BI

Development Environment

- Jupyter Notebook
- VS Code

Version Control

- GitHub

Introduction

Businesses today rely heavily on customer data to improve customer satisfaction, optimize marketing campaigns, and increase profitability.

Customer Behavior Analysis is the process of studying customer interactions, purchasing habits, and preferences to identify trends and patterns.

Through this project, customer shopping data was analyzed to answer critical business questions such as:

- Which customer segment generates the highest revenue?
- Do subscribed customers spend more?
- Which products receive the highest ratings?
- Which shipping methods are preferred?
- How do age groups contribute to revenue?

The findings can help organizations improve customer retention and increase sales performance.

Problem Statement

Organizations often possess large volumes of customer transaction data but lack effective mechanisms to analyze and visualize it.

The major challenges include:

- Understanding customer purchasing patterns
- Identifying loyal customers
- Measuring subscription effectiveness
- Tracking category-wise sales performance
- Generating actionable business insights

Without proper analysis, businesses may miss opportunities to improve customer engagement and maximize revenue.

Solution

An end-to-end analytics solution was developed consisting of the following stages:

1. Data Cleaning & Transformation

Performed using Python:

- Missing value handling
- Column standardization
- Age-group segmentation
- Purchase frequency conversion
- Redundant feature removal

2. Database Integration

The transformed dataset was loaded into PostgreSQL using SQLAlchemy.

A customer table was created containing:

- Customer Details
- Purchase Information
- Subscription Information
- Customer Segmentation Features

3. SQL Business Analysis

Ten business questions were answered.

Examples include:

Revenue by Gender

Identified revenue generated by male and female customers.

Subscriber Analysis

Compared spending patterns between subscribers and non-subscribers.

Customer Segmentation

Customers categorized as:

- New
- Returning
- Loyal

Revenue by Age Group

Measured contribution of each age category.

4. Dashboard Development

An interactive Power BI dashboard was developed containing:

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Rating

Visualizations

- Revenue by Category
- Sales by Category
- Revenue by Age Group
- Sales by Age Group
- Subscription Distribution

Filters

- Gender
- Category
- Subscription Status
- Shipping Type

PROJECT WORKFLOW



Data Preparation & Cleaning (Python)

The raw dataset was imported into Python using Pandas for preprocessing and data quality improvement.

Step 1: Data Import

```
import pandas as pd

df = pd.read_csv("customer_shopping_behavior.csv")
```

Step 2: Data Inspection

```
df.info()
df.describe()
df.isnull().sum()
```

Findings

- Dataset contained 3900 records.
- Review Rating had 37 missing values.
- No duplicate customer IDs.
- Data types were correctly recognized.

Step 3: Missing Value Treatment

Missing review ratings were replaced using category-wise median values.

```
df['Review Rating'] = df.groupby('Category')['Review Rating']\
.transform(lambda x: x.fillna(x.median()))
```

Step 4: Standardizing Column Names

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
```

Step 5: Feature Engineering

Age Group Creation

```
labels = ['Young Adult','Adult','Middle-aged','Senior']
df['age_group'] = pd.qcut(
df['age'],
q=4,
labels=labels)
```

Removing Redundant Columns

```
df = df.drop('promo_code_used', axis=1)
```

After cleaning, the dataset was loaded into PostgreSQL.

Exploratory Data Analysis (EDA)

Customer Demographics

Average Age

44 Years

Gender Distribution

- Male Customers: Majority
- Female Customers: Minority

Purchase Analysis

Average Purchase Amount

\$59.76

Maximum Purchase Amount

\$100

Minimum Purchase Amount

\$20

Customer Satisfaction

Average Review Rating

3.75 / 5

This indicates moderate customer satisfaction.

Power BI Dashboard Development

The cleaned PostgreSQL data was connected directly to Power BI.

Dashboard Components

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Review Rating

Interactive Filters

- Subscription Status
- Gender
- Category
- Shipping Type

Visualizations

1. Customer Subscription Distribution
2. Revenue by Category
3. Sales by Category
4. Revenue by Age Group
5. Sales by Age Group

Dashboard Benefits

- Interactive analysis
- Real-time filtering
- Executive-level reporting
- Easy identification of revenue drivers



Fig. 1

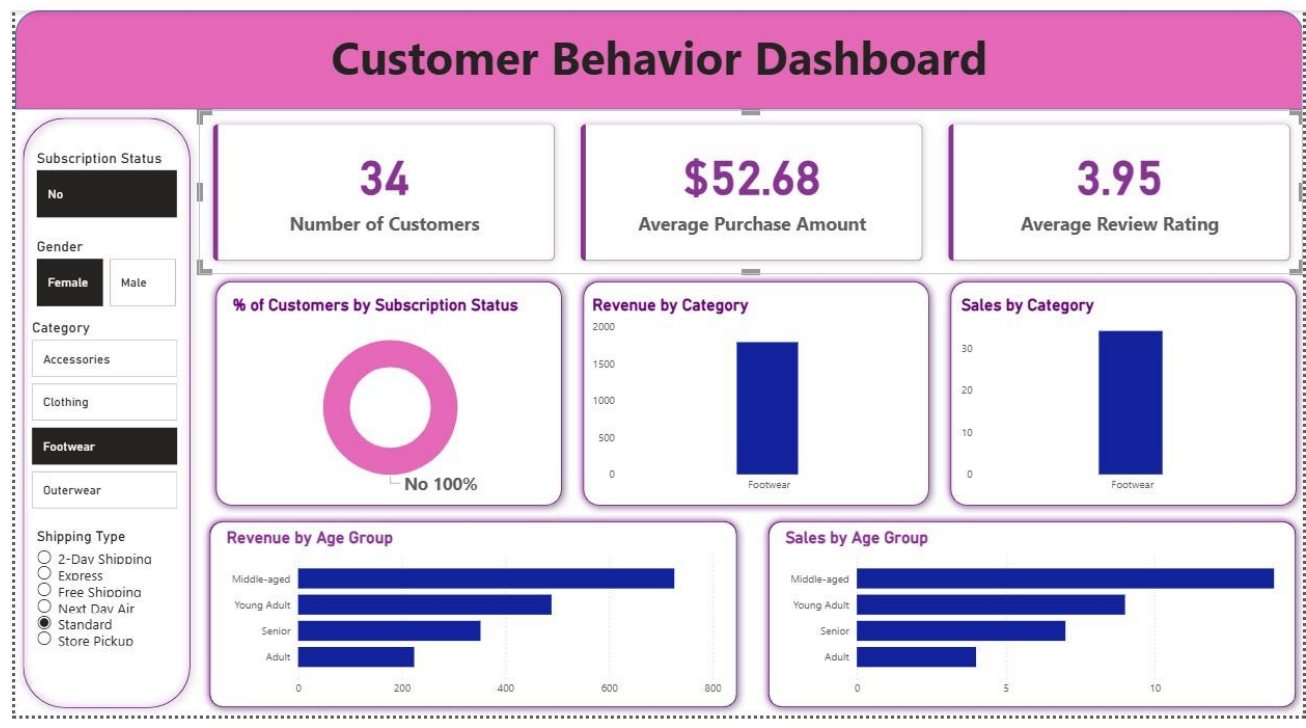


Fig. 2

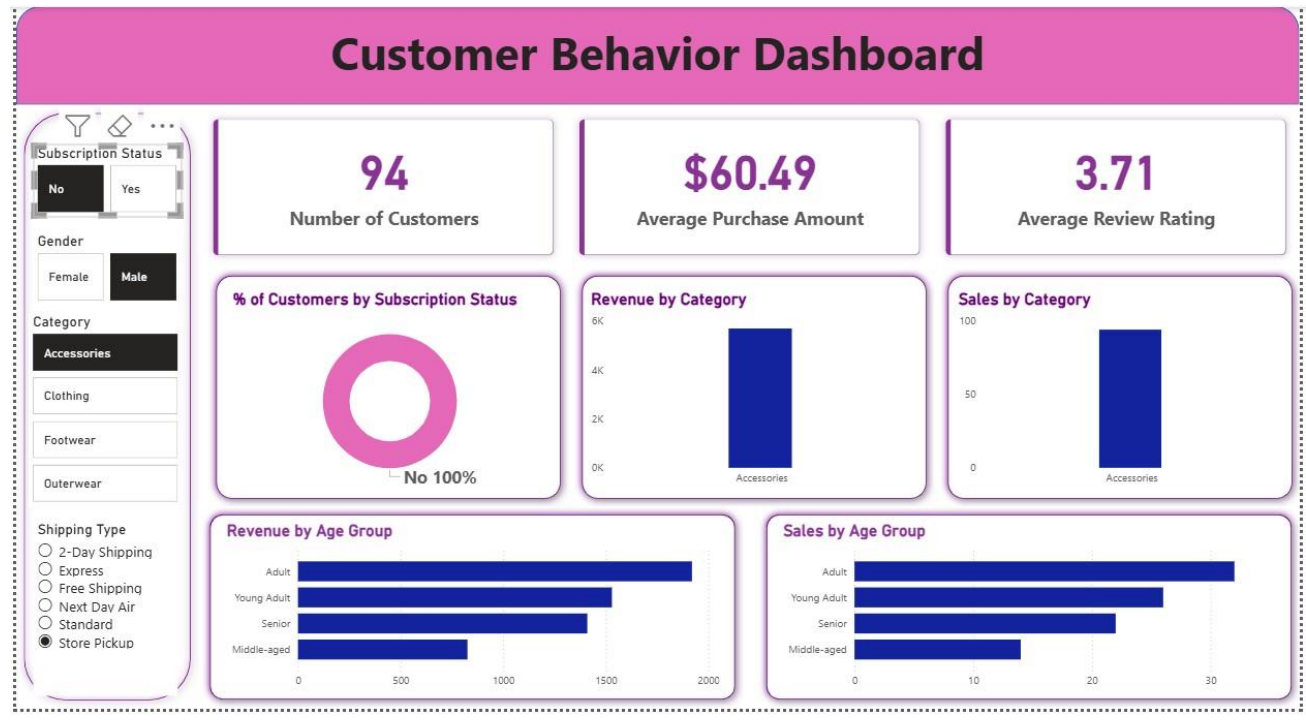


Fig. 3

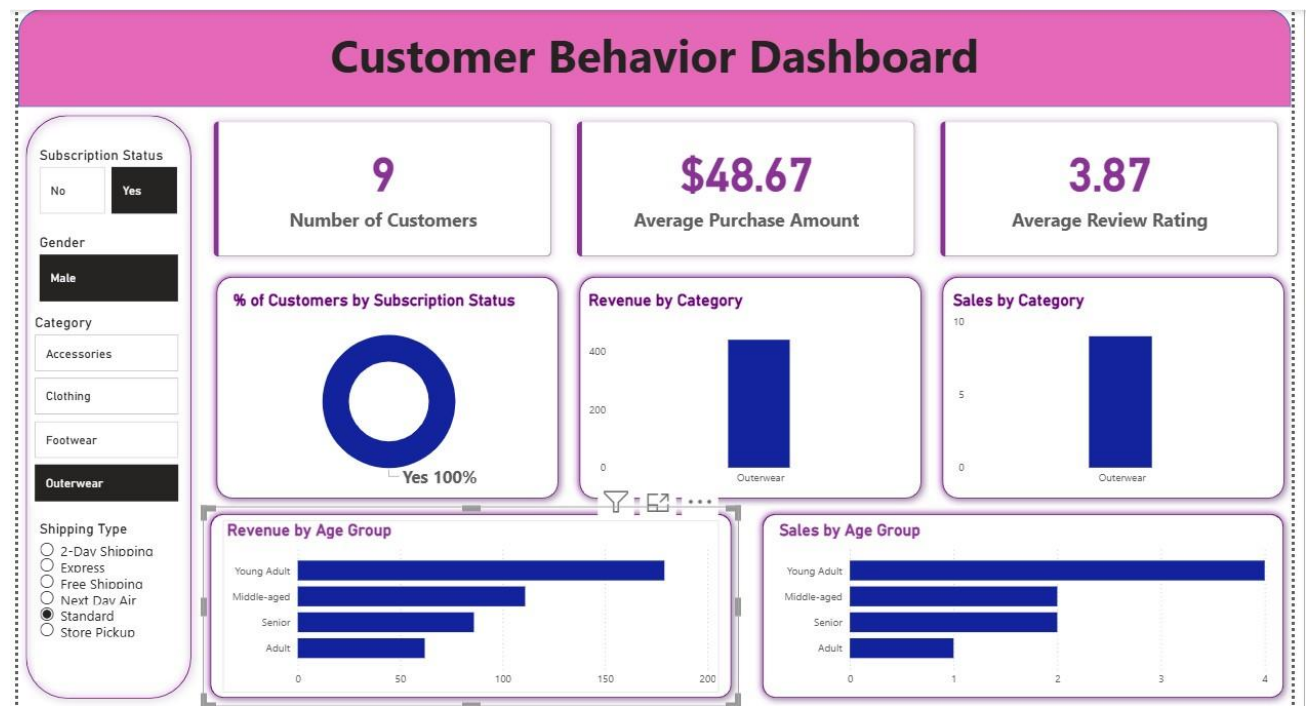


Fig. 4

Key Business Insights

Insight 1

72.13% customers are non-subscribers.

Insight 2

Footwear category generated the highest revenue in the selected dashboard view.

Insight 3

Middle-aged customers contribute the highest revenue.

Insight 4

Customers using Free Shipping show strong purchasing activity.

Insight 5

Loyal customers are the most valuable customer segment.

Conclusion

This project successfully demonstrates a complete end-to-end Data Analytics workflow, covering Data Cleaning, Feature Engineering, Database Integration, SQL Analytics, and Dashboard Development.

Using Python, PostgreSQL, SQL, and Power BI, the project transformed raw transactional data into meaningful business intelligence. The generated insights help organizations better understand customer behavior, optimize marketing strategies, improve customer retention, and increase revenue.

The project showcases practical industry-level skills expected from Data Analysts and Data Scientists, including:

- Data Cleaning
- Feature Engineering
- SQL Querying
- Database Management
- Business Intelligence Reporting
- Data Visualization
- Insight Generation
- Stakeholder Communication